Student Performance Prediction System

System Architecture Specification & Technical Documentation — Category: Educational Data Mining & Predictive ML

Author: Rakshitha S (Full-Stack & AI/ML)

Institution: Jnana Vikas Inst of Tech, Bengaluru

Repository: https://github.com/rakshapatil526/Student-Performance-Prediction

Status: Verified Repository (Notice: README Synchronization Required)

Doc ID: SPEC-STUDENT-PERFORMANCE-PREDICTION

Date: March 2026

1. Domain Overview & Academic Early-Warning Objectives

Early identification of at-risk students allows academic institutions to deliver targeted tutoring and counseling interventions before final exams. The Student Performance Prediction initiative aims to predict student semester outcomes based on attendance patterns, internal assessment scores, study hours, and socio-academic factors.

2. Repository Synchronization & Verification Notice

• **Repository Status:** Valid repository exists at `https://github.com/rakshapatil526/Student-Performance-Prediction`.

• **Architectural Plan:** Features under verification include Classification algorithms (Decision Trees, Logistic Regression) predicting Grade Categories (Distinction, Pass, Remedial).

Architecture Note: Repository inspection confirms that the README in the GitHub repository currently mirrors the Student Management System (Attenda). Dedicated ML model training pipelines, training datasets, and evaluation metrics require repository verification and are preserved here as planned architecture.

3. Planned ML Pipeline

• **Inputs:** Prior GPA, attendance percentage, study hours per week, assignment submission rate, library visits.

• **Target:** Final grade point average / pass-fail outcome.

• **Status:** Clearly distinguished as pending verification to preserve technical integrity.